

JMod Outputs

List of all output files produced by a JMod search and all columns that can be found in those output files.

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All columns

Description of columns in JMod output files

coeff	Coefficient estimated for the precursor
spec_id	MS2 spectrum ID where the precursor was found
Ms1_spec_id	Closest MS1 spectrum to the MS2 spectrum
seq	Precursor sequence
z	Precursor charge
window_mz	Center of the MS2 isolation window
rt	Retention time of the MS2 spectrum
num_lib	Number of library fragments matched
frac_lib_int	Fraction of the library intensity matched
frac_dia_int	Fraction of the observed intensity matched to the precursor
mz_error	Relative MS1 m/z error
rt_error	Retention time error (empirical RT - library RT)
frac_int_matched	Fraction of the total spectrum intensity with any match (any precursor)

frac_int_pred	Fraction of the total spectrum intensity predicted by all precursors
spec_r2	Pearson correlation for the predictions and observed spectrum
prec_r2	Pearson correlation for the predicted precursor and matched spectrum peaks
prec_r2_uniq	Pearson correlation for the predicted precursor and uniquely matched peaks
frac_int_uniq	Fraction of library intensity uniquely matched
frac_int_uniq_pred	Fraction of total spectrum intensity uniquely predicted
hyperscore	Sum of matched intensities times factorial(#b ions) times factorial(#y ions)
b_counts	Number of b ions matched
y_counts	Number of y ions matched
longest_y_ions	Index of largest y ion matched
scribe_scores	Scribe score as described here: https://pubs.acs.org/doi/abs/10.1021/acs.jproteome.2c00672
max_unmatched_residuals	Maximum residual for unmatched peaks, normalized and log-transformed.
max_matched_residuals	Maximum residual for matched peaks, normalized and log-transformed.
gof_stats	Goodness-of-fit score for each precursor (log2 of residuals/fitted).
manhattan_distances	Manhattan distances between precursor peaks and observed peaks. (normalized and log-transformed)
fitted_spectral_contrasts	Spectral contrast angles for each precursor
frac_int_matched_pred	Fraction of the matched spectrum intensity predicted by all precursors
frac_int_matched_pred_sigcoeff	Fraction of spectrum intensity predicted by precursors with coeff > 1
cosine	Similarity between predicted spectra and observed intensities
mz	Precursor m/z
tic	Total ion current
file_name	Name of the file searched

protein	Protein associated with the precursor
is_decoy	Is this a decoy peptide
n_scans	Number of scans where that precursor was identified.
manhattan_distances_nearby_max	Maximum manhattan_distances of nearby scans
max_matched_residuals_nearby_min	Minimum max_matched_residuals of nearby scans
gof_stats_nearby_min	Minimum gof_stats of nearby scans
scribe_scores_nearby_min	Minimum scribe_scores of nearby scans
smoothness	The smoothness of coeffs for that precursor
stripped_seq	Amino acid sequence without tags or mods.
pep_len	Peptide sequence length
sq_rt_error	Square root of the rt_error
sq_mz_error	Square root of the mz_error
untag_seq	Peptide sequence without the tag (contains mods)
untag_prec	Untag_seq and charge combined
channels_matched	Number of channels from the plex set matched
median_[feature]_stats	Median of [feature] across channels
diff_[feature]_from_median	Difference between channel [feature] and the median [feature] value across channels
frac_shared_intensity	Fraction of shared intensity between channels
channel	Channel name of the precursor
silac_channel	SILAC channel
med_frag_error	Median absolute mz error of the matched fragments
n_corr_scans	Number of scans used in correlation calculation

n_corr_frgs	Number of fragments used in correlation calculation
mean_frag_corr	Mean correlation between fragments
median_frag_corr	Median fragment correlations between fragments for the same precursor
max_frag_corr	Maximum fragment correlations between fragments for the same precursor
min_frag_corr	Minimum fragment correlation between fragments for the same precursor
std_frag_corr	Standard deviation of fragment correlations
mean_top3_frag_corr	Mean of the top 3 fragment correlations
frac_corr_above_0p5	Fraction of fragments with correlation above 50%
mean_frag_mean_corr	
top_[n]_frag_mean_corr	Mean of top [n] fragment correlations
top_[n]_frag_sum_corr	Sum of top [n] fragment correlations
top_[n]_pair_corr	Top [n] correlations between fragments
mean_prec_frag_corr	Mean precursor to fragment correlation
max_prec_frag_corr	Maximum precursor to fragment correlation
PredVal	Raw score from the target-decoy classifier (1 is target)
Qvalue	Q-value of the precursor
BestChannel_Qvalue	Best q-value of the precursor plex set
plexfitMS1	Fitted MS1 coefficient at max
plexfitMS1_p	Correlation of theoretical MS1 isotope distribution with observed peaks
plexfittrace	Fitted MS1 coefficients of elution trace
plexfit_ps	Correlation of theoretical MS1 isotope distribution with observed peaks for each fit in elution trace
plexfittrace_spec_all	All MS1 spectra used for MS1 fitting
plexfittrace_all	All fitted MS1 coefficients

plexfittrace_ps_all	Correlation of theoretical MS1 isotope distribution with observed peaks for each fit
plex_Area	Area under fitted MS1 coefficients
ms1_apex_scan	MS1 apex scan selected for ms quantitation
ms1_cor	Correlation between MS2 coefficients and monoisotopic trace
traceproduct	Product of MS1xMS1 correlations
iso_cor	Correlation between theoretical and observed MS1 isotopes
MS1_Int	Observed intensity of MS1 monoisotopic peak at max
all_ms1_specs	MS1 spectra searched for monoisotopic peak
MS1_Area	Area under elution trace of MS1 monoisotopic trace
all_ms1_iso[n]vals	Observed MS1 intensities for the [n]th isotopic peak (0 being the monoisotopic peak)
last_aa	Last amino acid in the peptide sequence
seq_len	Length of the amino acid sequence
run_chan	Sample run and channel concatenated
Protein_Qvalue	Protein q-value
frag_names	Names of fragments matched
frag_errors	m/z errors of fragments matched
frag_mz	Theoretical m/z of fragments matched
frag_int	Library intensity of fragments matched
obs_int	Observed intensity of fragments matched
unique_frag_mz	Fragments that uniquely matched a peak
unique_obs_int	Intensities for fragments that uniquely matched a peak

Output Files

List of output files produced by JMod with a description of them and what columns they contain.

filtered_IDs.parquet

List of precursors filtered at 1% FDR. Each row represents a single precursor [sequence, charge]. Contains the following reduced number of columns for fast analysis:

- stripped_seq
- z
- untag_prec
- file_name
- channel
- is_decoy
- Qvalue
- Protein_Qvalue
- PredVal
- protein
- BestChannel_Qvalue
- plex_Area
- seq
- silac_channel
- untag_seq
- rt
- mz

outputs/all_IDs_filtered.parquet

List of all precursors, including decoys, considered during the search. Each row represents a single precursor [sequence, charge]. Contains a reduced number of columns for fast analysis (same as above).

filtered_IDs.csv

List of precursors filtered at 1% FDR. Each row represents a single precursor [sequence, charge]. Contains all columns

outputs/all_IDs.csv

List of precursors not filtered at 1% FDR. Each row represents a single precursor [sequence, charge]. Contains all columns

outputs/decoylibsearch_coeffs.parquet

List of all candidate peptide-spectrum matches considered during the search, including decoys. Each row is a unique [sequence, charge, scan].